

Design of a Hydraulic System for a Lorry Crane



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1 Introduction

The aim of this task is to illustrate some of the problems that engineers can encounter when constructing hydraulic systems for mobile machinery. The assignment is intended to provide training in problem solving related to hydraulic system design.

1.1 Practical Information

To pass the assignment, the following instructions should be followed:

I. Signing up:

The task should be performed in groups of 2-3 students. Sign-up is made on Lisam. If you do not have anyone to work with, you can either ask to join a group with one or two members (it is good if the partner(s) take approximately the same courses as you), or sign up for an empty group and hope that someone will contact you for collaboration. Each group number has a specific set of parameters (found in a separate document).

II. Quizzes:

There are two quizzes on Lisam that you should do to make sure that your calculations are correct. The parameters used in the quiz are NOT connected to your group-specific parameter set. Use the quizzes to verify your own calculations.

III. Presentation and preliminary report:

You should submit your report on Lisam and prepare a 10-minute presentation about it. The presentation should focus on your results and you should explain why the plots look as they do and justify the choices you have made.

There will be several presentation sessions. You sign up for a session on Lisam. For each session, one group will be randomly selected to present their work to the other groups. After the presentation there will be a discussion. Active participation is required to pass.

The intention of this session is to make sure that everyone has an understanding of the task, and that they do not simply rely on a friend (human or not).

IV. Final report:

After the presentation, all groups have the opportunity to revise their reports. If you do not think that you need to make any changes, you should submit the preliminary report again.

V. Deadlines and submission:

The following list summarizes the submissions and tasks mentioned above. All dates can be found on Lisam.

- Quiz: Design Task – Task 1–3b
- Quiz: One of the system quizzes
- Preliminary report
- Presentation (no submission)
- Final report

1.2 Report Requirements

The front page must include the group number as well as the full names and LiU-IDs of all group members.

The report is expected to be well structured and well written. The target audience should be another student in this course. Therefore, it is not important to explain the task itself and other background information. However, full solutions of calculations and assumptions should be clearly stated and justified. Figures should be readable and understandable. A list of all used parameter values should be included. In short, make sure that all tasks and questions in section 3 are addressed in the report.

The final reports will be corrected as soon as possible after the last day of submission. If a report is failed, you only have one more chance to submit the report. It is recommended to talk to a teacher before

handing in any resubmission. A delayed or failed resubmission will not be corrected until the next re-exam period (i.e. in January).

It is **not allowed** to copy solutions from other groups. To avoid confusion, please read the university guidelines on plagiarism: <https://liu.se/en/article/plagiering-upphovsratt>. In addition, it is **not allowed** to use AI-generated material that you do not understand. AI may be used as a discussion partner and as a tool for specific subtasks, but **not** to write entire reports. You must be able to explain all material included in your report.

2 Design of a Hydraulic System for a Lorry Crane

In this task, you shall design a hydraulic system for controlling two arms on a lorry crane. The hydraulic system shall include a supply unit (pump unit) and valves for controlling the two cylinders shown in red in Figure 1.

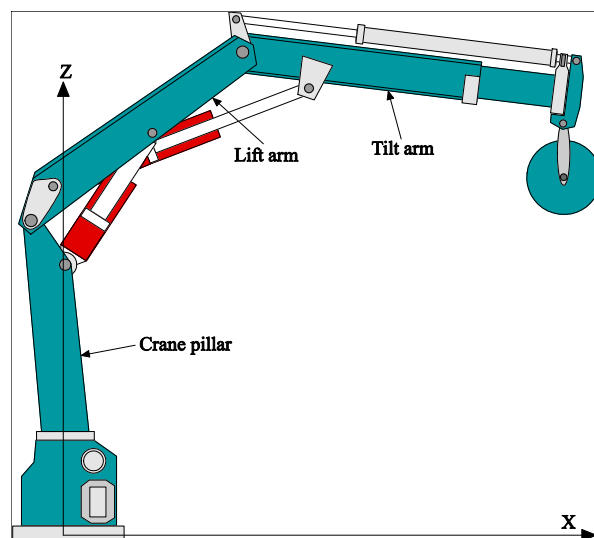


Figure 1: A lorry crane and its main components.

The crane is to be analysed and designed in several steps. The hydraulic system shall first be dimensioned for a worst-case scenario that the crane must be able to handle. With the help of a drive cycle representing a typical use case of the crane, one of the following system solutions is then to be selected for the crane:

- Load-sensing system (LS)
- Constant-flow system with a fixed pump (CF)
- Constant-pressure system with a variable pump (CP)

The hydraulic system shall be designed with respect to energy consumption, cost, and safety aspects. The cylinders must be double-acting for controllability reasons. Pressure losses in hoses and valves shall be considered, which means that the cylinder inlet pressure (load pressure) must be lower than the maximum system pressure ($p_{s,max}$). Efficiency models for the pumps shall be taken from Appendix 1. Once one of the system solutions above has been chosen, the energy efficiency of the chosen system configuration is to be evaluated for the drive cycle.

2.1 Crane geometry and load specification

Figure 2 shows the crane with the parameters and variables needed to describe the geometry and the drive cycle.

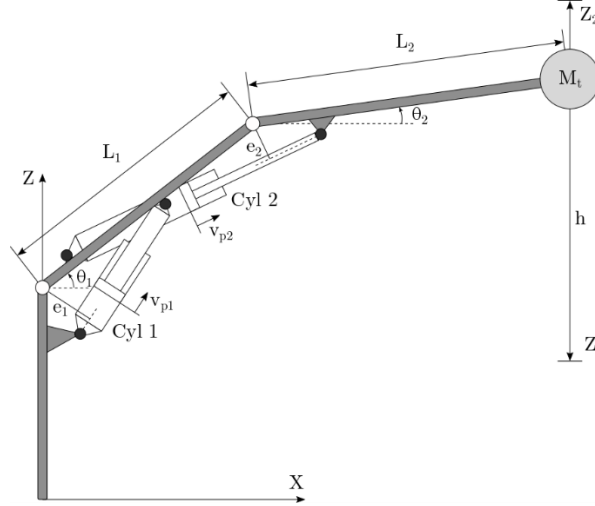


Figure 2: Crane structure and parameters.

The mass M_t is assumed to be concentrated at the crane tip. The mass of the arms can be neglected. The lifting arm has the length L_1 and the tilting arm has the length L_2 . Both arms have constant lengths. The crane pillar has a height of 3 m.

In addition to the parameters above, the crane geometry is defined by the angles of the arms relative to the horizontal plane, θ_1 and θ_2 , and the cylinder lever arms, e_1 and e_2 . The velocities of the cylinder pistons are v_{p1} and v_{p2} , respectively.

2.2 Worst-case scenario

To fulfil the worst-case scenario, the crane must be able to lift a load with the total mass $M_{t,wc}$. The cylinder forces depend on the crane geometry, and the maximum force in each cylinder occurs at the angles and lever arms given in Table 1.

Table 1: Crane angles and lever arms corresponding to the maximum cylinder forces.

Parameter	Cylinder 1	Cylinder 2
$\theta_{1,wc}$ [rad]	1.2	0.35
$\theta_{2,wc}$ [rad]	0.0	0.0
$e_{1,wc}$ [m]	0.14	0.48
$e_{2,wc}$ [m]	0.57	0.17

Furthermore, the cylinders must be able to operate at higher speeds than those required by the drive cycle. A group-specific parameter, $q_{r,wc}$, is therefore given. It is defined as the ratio between the maximum flow required during the drive cycle and the flow required for the worst-case scenario:

$$q_{r,wc} = \frac{q_{dc,max}}{q_{wc}}. \quad (1)$$

2.3 Drive cycle

The drive cycle consists of a vertical movement of the mass M_t over a distance h from Z_1 to Z_2 and back to Z_1 , as shown in Figure 2. The relative cylinder velocities that describe the cycle are shown in Figure 3. Similarly, the cylinder lever arms and the crane arm angles during the cycle are shown in Figures 4 and 5, respectively. Note that the time axis is normalised in all graphs. The vectors for each curve are available in a MATLAB file on Lisam. However, for the calculations, you may use the calculation program of your choice.

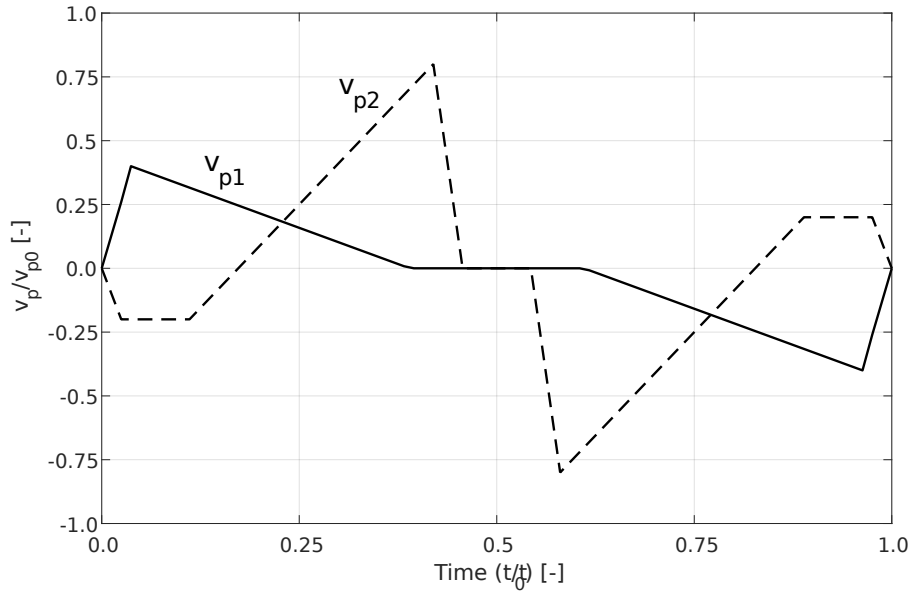


Figure 3: Relative cylinder piston velocities versus time during the drive cycle.

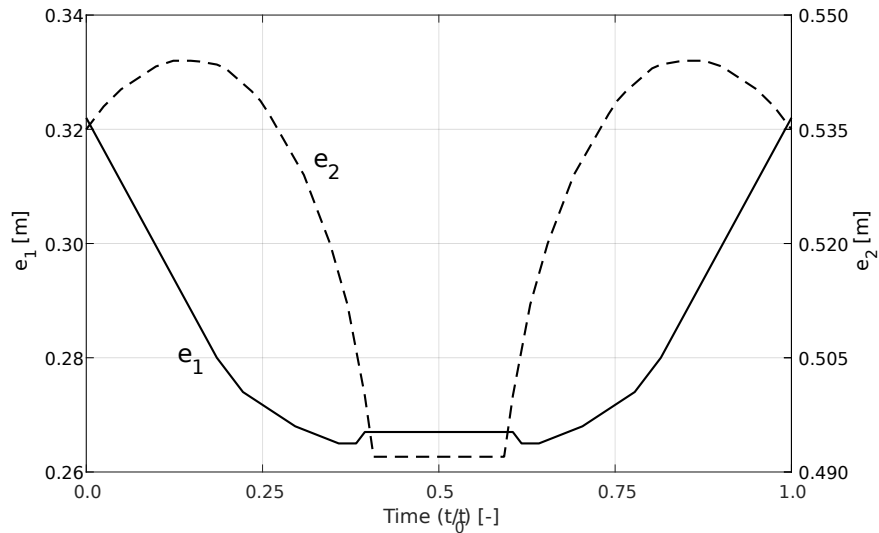


Figure 4: Cylinder lever arms versus time during the drive cycle.

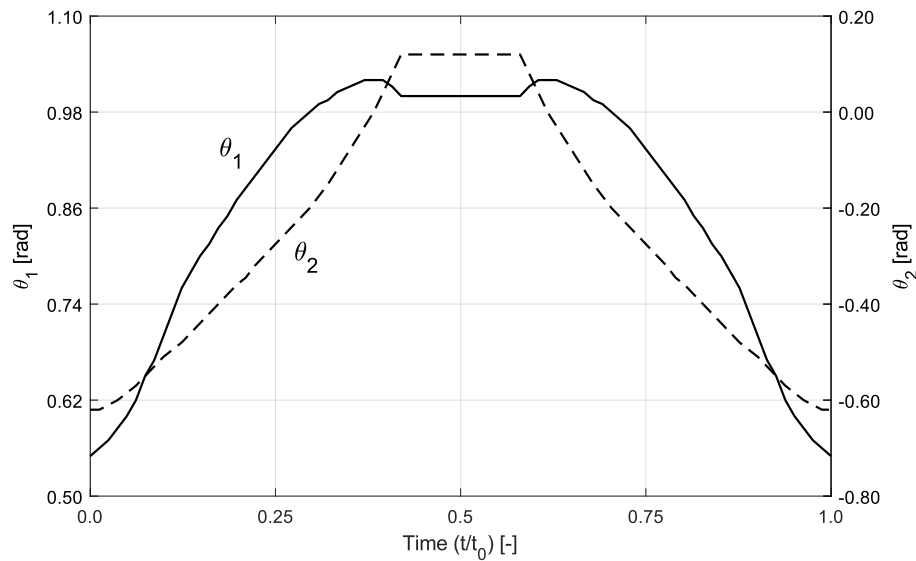


Figure 5: Crane arm angles versus time during the drive cycle.

2.4 Parameter values

The parameters required to carry out the calculations are divided into two sets: one that is common to all groups and one that is unique to each group. The common parameters are provided in this document and some can also be found in the MATLAB file *vectors.m* in the course documents section on Lisam. The file *vectors.m* contains the parameters describing the crane geometry for the worst-case scenario and the crane pillar height (L_0). It also contains vectors of the quantities needed to describe the drive cycle (i.e. the graphs in Figures 3–5). Note that the velocity vector is normalised with respect to v_{p0} (see below). The group-specific parameters can be found in the file *Data_tmhp02.pdf* on Lisam. The following parameters are provided (see *Data_tmhp02.pdf* for numerical values):

- Maximum total load mass: $M_{t,wc}$ [kg]
- Total load mass: M_t [kg]
- Maximum system pressure: $p_{s,max}$ [MPa]
- Maximum pump flow ratio: $q_{r,wc}$ [-]
- Lifting arm length: L_1 [m]
- Reference cylinder velocity: v_{p0} [m/s]
- Tilt arm length: L_2 [m]
- Cylinder efficiency: $\eta_{c1} = \eta_{c2} = \eta_c$ [-]
- Reference time for the drive cycle: t_0 [s]

Note that the values of some parameters required for the calculations are not given explicitly and must therefore be determined based on reasonable assumptions. This is common practice in engineering, as all necessary data are seldom available during the design process. Remember to *clearly* state and justify your assumptions in the report.

3 Tasks

You are supposed to design a safe and functional hydraulic system for the crane and analyse its performance. To facilitate the assessment, the task is divided into several sub-tasks. The plots stated below should be accompanied by full calculations, explanations of the results, and any assumptions made. Note that there might be several approaches to solve these tasks.

1. Analyse the crane's movement during the drive cycle:
 - a. *Calculate* and plot the Z-position of the crane tip as a function of time during the drive cycle. Also, plot the crane tip position as X vs Z. *Are your graphs realistic? Why?*
 - b. *Calculate* and plot how the external forces acting on the two cylinders vary during the drive cycle.
2. Dimension cylinders for the worst case:
 - a. *Calculate* the minimum allowed cylinder diameters, D_1 and D_2 , to be able to lift the *maximum* load $M_{l,wc}$. Choose appropriate cylinders. *Remember to consider the cylinder efficiency and pressure drops.*
 - b. Present your calculated and chosen parameters. Clearly state which cylinders you have chosen (brand, type, size). The stroke lengths of the cylinders do not need to be considered.
3. Choose a suitable valve-controlled system for the crane (use the drive cycle):
 - a. *Calculate* and plot how the pressures (both chambers) in the two cylinders as well as the minimum required pump pressure vary during the drive cycle. In the same graph, include the maximum allowed system pressure (i.e. in total 6 curves in one graph).
 - b. *Calculate* and plot how the flow required from the pump by each cylinder varies during the drive cycle. Include the total required pump flow and the maximum pump flow in the same graph (i.e. in total 4 curves in the same graph).
 - c. Choose a valve-controlled hydraulic system (LS, CP or CF) for the crane. Motivate your choice by discussing the aspects *cost*, *driveability* and *efficiency*. *The results from 3a and 3b can give guidance in your decision.*
 - d. Make a circuit diagram of your proposed hydraulic system for the crane. Remember to include necessary components to ensure safety and driveability (for instance, some kind of load-holding valves are required for the cylinders).
4. Choose a pump (*remember to consider the flow demand for the worst case*):
 - a. Select a pump that is suitable for your specific system and flow demands from one of the suppliers listed in *Hydraulic Components.pdf*.
 - b. Present relevant data (pump type, displacement, maximum speed, operating speed, maximum pressure, etc.) clearly. Note that your maximum system pressure still is limited to your individual parameter due to other components and hoses.
5. Investigate how the system you have chosen performs during the drive cycle:
 - a. Present how the pump pressure in your system varies during the drive cycle (it is ok to refer to a curve in Task 3a).
 - b. Present how the pump flow in your system varies during the drive cycle (it is ok to refer to a curve in Task 3b).
 - c. Plot how the pump displacement setting varies in your system during the drive cycle. In another graph, plot how the pump efficiencies vary in your system during the drive cycle (3 curves: volumetric, hydro-mechanical and total pump efficiency). Use the efficiency models on p. 9 in the formula book.

- d. Plot the input power to the pump and the output power from the cylinders during the drive cycle in the same figure. How do you consider a “*negative output power*” component?
 - e. Some general reflections: Are the losses high/low? Why? How would it change if you had used another system type? Would you recommend a crane manufacturer to use your system?
6. Dimension a cooler for your system, considering a full cycle:
- a. Some guiding questions to help you with the calculations: *What is the net useful mechanical work output for a full cycle (lifting and lowering of the mass)? No calculations are needed. What does this imply regarding the relation between input power and losses?*
 - b. *Calculate* the required average cooling power (the simplest approach might be to make use of the information gathered in Task 6a).

4 General assumptions

The following assumptions can be made during the calculations:

1. The weight of the crane can be ignored.
2. Acceleration forces can be ignored.
3. The gravitational constant is 9.81 m/s^2
4. The low-pressure side in the cylinders is always 1.0 MPa.
5. The tank pressure can be ignored.
6. The leakage flow in valves and cylinders can be ignored.
7. The dynamic viscosity of the oil, η , can be assumed to be $\eta = 0.03 \text{ N s/m}^2$.
8. You are supposed to use the efficiency models on p. 9 in the formula book. The following parameters should be used:
 - $c_v = 1.3 \times 10^{-9} \text{ [rev]}$
 - $k_p = 5.4 \times 10^{-2} \text{ [-]}$
 - $k_v = 6.5 \times 10^{-5} \text{ [1/rev]}$
 - $k_e = 0.3$ (for variable-displacement pumps)

You will need to make some additional assumptions.